HRIDAYA SHRESTHA

UNDERGRADUATE COMPUTER SCIENCE STUDENT

HOW TO REACH ME

Email : ridaya350@gmail.com
Location : Chandragiri-06, Kathmandu, Nepal
Contact no. : +977 9825805154

LinkedIn : linkedin.com/in/ridayashrestha/
Github : github.com/cherrrydo

EDUCATIONAL BACKGROUND

Bachelor of Computer Science and Information Technology - BCS.IT
Institution : Himalayan College of Management (HCM)
University : Infrastructure University of Kuwala lumpur
Current : 4th semester
Website : hcm.edu.np/program/bachelor-of-computer-science-hons

Higher International Diploma in Computing (EQF)
Major : Software Development
Institution : Dursikshya Education Network
Awarding Body : Scottish Qualification Authority (SQA)
Website : sqa.org.uk/sqa/81334.html

12th National High School Diploma
Major : Computer Science
Institution : G.S College of Management
Awarding Body : Tribhuwan University (T.U)
GPA : 3.30
Website : https://tu.edu.np/

TECHNICAL SKILLS

Frontend Development
- HTML5/CSS3, JavaScript (ES6), TailwindCSS

Backend Development
- Python, Django

Database Management System
- MySQL, Relational Databases, MongoDB

Version Control & Collaboration
- Git
- Github

Other Tools & Technologies
- JavaScript, RESTful APIs
- Seaborn, Pandas, Matplotlib, jQuery, Tkinter
- Microsoft Excel, Data Analysis & Visualization, Canva

PROFESSIONAL SUMMARY

I'm a Computer Science student looking for an internship where I can apply what I've learned in class and get real-world experience in the IT industry. Excited to work on innovative projects, learn from professionals, and build my skills in areas like software development, networking, and cybersecurity.

TECHNICAL PROJECTS

Full-Stack Python D-jango Ecommerce Platform
Capstone Project

- Featuring **product listings,** a **shopping cart**, and **secure checkout**.
- Integrated **AI tools** to enhance **product recommendations.**
- Focused on **user experiences** and **search functionalities,** making the platform more user-centric.
- Implemented **user authentication** and **authorization**, allowing users to **register, login,** and **manage** personal information, as well as order history.
- Utilized **Git** for **version control**, ensuring smooth **collaboration** and proper **tracking** of changes throughout the **development cycle**.
- **Link:** https://github.com/cherrrydo/nicecart-django.git

Enterprise Network Design & Implementation for Business Expansion
Dursikshya Education Network Pvt Ltd | Scottish Qualification Authority

- Designed and implemented network topology with **redundancy** at every layer using two routers and two multilayer switches.
- Configured connections to two **ISPs**, ensuring network redundancy and uninterrupted internet connectivity.
- Divided the network into multiple VLANs for each department, performing **subnetting** on the base network (**172.16.1.0**) for efficient IP address management.
- Configured **OSPF** routing protocol for dynamic routing between routers and multilayer switches to ensure optimal network performance.
- Implemented inter-VLAN **routing** on multilayer switches to enable seamless communication between different department VLANs.
- Set up a DHCP server for **dynamic IP address** assignment with **DHCP** helper addresses to ensure cross-VLAN IP allocation.
- Secured network access by configuring **SSH** on all routers and switches for **remote access** management.
- Applied **port security** on Finance and Accounts department switches using sticky MAC addresses and violation mode set to shutdown.
- Configured **wireless** access points for department-specific wireless connectivity while maintaining secure network segmentation.
- Implemented **NAT (PAT)** and **ACLs** for secure internet access and **traffic flow control**.
- Conducted thorough **testing** and verification of network functionality, ensuring proper inter-VLAN communication and redundancy.

Problem Solving & Optimization

- Debugging, Code Optimization, Technical Support, Ethics

Student Performance Data Analytics Project
Dursikshya Education Network Pvt Ltd | Scottish Qualification Authority

- Collected and preprocessed a **dataset** containing student **demographics**, **academic records**, and related features.
- Handled **missing values**, **removed duplicates**, and **transformed variables** for analysis.
- Conducted EDA using Python libraries (**Pandas, Numpy, and Seaborn**) to discover trends, correlations, and patterns in the data.
- Created visualizations such as **histograms**, **scatter plots**, and correlation matrices to interpret relationships between variables.
- Preprocessed the dataset by **encoding** categorical variables and **scaling** numerical features.
- Split data into **training and testing** sets to evaluate model performance.
- Built a **linear regression** model to predict student performance based on multiple factors.
- Evaluated the model using **metrics** like mean squared error **(MSE) and R-squared.**
- Visualized the **predictions** (y_pred) versus actual values (y_test) to assess model **accuracy**.
- Generated various **data visualizations** to present findings and insights effectively.

Advance Database Project for E-commerce Platform
Dursikshya Education Network Pvt Ltd | Scottish Qualification Authority

- Designed and implemented a **normalized** SQL-based database for an e-commerce platform, ensuring data integrity and optimized performance.
- Developed an **entity-relationship diagram (ERD**) and data dictionary to structure and document database entities.
- Created **SQL scripts** for table creation, data population, and report generation, including **sales** summaries, **product** inventory, and **customer** order history reports.
- Implemented **order lifecycle management**, product reviews, and multi-tenant support to enhance database functionality.
- Utilized **foreign keys**, **constraints**, and indexes to maintain data consistency and improve query performance.

Blog Website Development & Bug Fixing (Django)
Self

- **Optimized Django application** by identifying and fixing critical bugs, improving performance and user experience.
- **Resolved database issues** related to model relationships, migrations, and query optimizations for better efficiency.
- **Fixed authentication and authorization bugs** to enhance user security and access control.
- **Improved front-end integration** by troubleshooting template rendering issues and fixing broken UI components.
- **Refactored backend logic** to improve code maintainability and scalability.
- **Debugged and resolved API issues**, ensuring seamless communication between frontend and backend.
- **Enhanced SEO & URL structure**, fixing routing issues for better search engine visibility.
- **Deployed and tested fixes** in a staging environment before going live to ensure stability.

For more projects: github.com/cherrrydo